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Brown Dwarfs

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1 Introduction

What is a Brown Dwarf ? Defining a brown dwarf isn't straightforward because it sits at the boundary between the largest planets and the smallest stars. A brown dwarf is often described as a failed star or a sub-stellar object. Unlike stars, brown dwarfs lack sufficient mass to sustain the nuclear fusion reactions in their cores, which are necessary for them to shine like stars. One commonly used criterion to distinguish brown dwarfs from planets and stars is based on their ability to fuse deuterium in their cores during their early formation stages (although the burning of deuterium has a negligible impact on stellar structure and evolution). We define a “*low-mass star*” as having a mass between $\sim 0.2M_{\odot}$ and the hydrogen burning mass limit ($\sim 0.075M_{\odot}$ or $70 - 80M_J$). This is also the lower mass limit for main-sequence (M-S) stars. All objects below this mass range are considered *brown dwarfs*. The minimum mass for brown dwarfs is that of deuterium fusion (at $0.01M_{\odot}$ or $13M_J$). The theory of the formation of brown dwarfs equates them to the formation of stars, through the direct collapse of an interstellar cloud. During this process, material with high angular momentum forms a circumstellar disk, in which dust can aggregate into planetesimals which, in turn, can fuse to form rock nuclei. When a nucleus reaches sufficient mass, it can begin to grow rapidly, attract surrounding gas, and become a gas giant around the brown dwarf. Brown dwarfs are usually considered relatively young objects, as the process of brown dwarf formation is often associated with star formation. We will see that there will be different environments in which to find brown dwarfs and depending on the environment, associate a certain age. Since we are examining the formation and early evolution of low-mass stars and brown dwarfs, we focus on objects with ages $\lesssim 100$ Myr, which is the time-scale for low-mass stars to approach the main sequence. In summary, the definition of brown dwarfs should have a basis more similar to the definition of stars (based on internal physics) and since the critical difference between stars and planets is that stars experience nuclear fusion, then we can place a mass limit on BD based on deuterium fusion.

Where are the brown dwarfs? There are several methods to search for brown dwarfs. Studies on the formation and initial evolution of low-mass stars (LMS) and BDs have been made possible by surveys like *2MASS*, that discover these objects, particularly at a young age. The targets of these surveys consisted of the closest young associations, open clusters and star-forming regions. The three main ways to detect brown dwarfs are through direct imaging, multiple systems (Doppler effect) or looking for newborn brown dwarfs. However at early age BDs occupy the same temperature and luminosity region as very low mass stars (VLMS) and this will make it difficult to verify that they are actually BDs. Brown dwarfs can be observed in the infrared ($1-2 \mu\text{m}$) but also be detected in the optical ($0.5 \mu\text{m}$). It is actually easier to identify cool dwarfs than many other classes of astronomical sources due to the nature of their spectral energy distributions (SEDs), which arises from strong molecular absorption bands (see Sect. (3)). In 1995 there was the first successful application of the **lithium test** (see Sect. (2)) for substellarity. The search was aimed at the Pleiades cluster, where the object **PP1 15** has a mass just below the sub-stellar limit. Consequently, all weaker members of the Pleiades should be BDs. Indeed, faintest objects in the cluster are expected to be brown dwarfs rather than stars, since their mass is just below the sub-stellar limit. This does not mean that all the dimmer objects in the Pleiades are necessarily BDs, but that a good portion of them are likely to be. A few months later, the discovery of an object of this type was announced: **Teide 1**. Then, the discovery of a very faint and cold companion around an M-type (Gl 299A) star was announced. The object is **Gl 229B** and it has been the definitive proof of the existence of BDs.

2 Lithium Test

Since the difference between BDs and VLMS (low-mass stars) lies in the nuclear behavior of their cores, it is natural to look for a nuclear test of substellarity. In addition to verifying sub-stellar status, observations of lithium can be used to assess the age of stars in clusters. In simplest terms, stars will burn lithium in a little over 100 Myr at most, whereas most BDs will never reach the core temperature required to do so. The lithium test for a brown dwarf involves measuring two things: the lithium content and the spectral type or luminosity. But you may also need to know (or assume) something about the age of the object. One thing needs to be clarified immediately: the Li-test is not a completely unambiguous indicator of whether an object of unknown age is a star or a brown dwarf. Both stars and brown dwarfs can have spectral types M6.5-L2, but a star would be older at the same spectral type. The issue is: if we observe an object that does not contain lithium, how do we understand that it is a brown dwarf or a star that has already finished burning lithium? The minimum lithium burning temperature was never reached in the cores of objects below about $60 M_J$. One really needs to know the age of an object in order for its luminosity or T_{eff} to lead to even a crude mass estimate and hence the Li-test is pursued. Lithium begins to burn at temperatures above $3 \cdot 10^6$ K. The time it takes for a pre main sequence star (or some brown dwarfs) to reach this temperature depends on the mass. It takes about 20 Myr at $0.2 M_\odot$, about 120 Myr at $0.075 M_\odot$, and if the object is below $0.06 M_\odot$, he never gets rid of his Li. However, if it was possible to measure the luminosity (or spectral type or temperature) of an object and know whether it has lithium or not, then this provides a constraint on both mass and age. Figure (1) shows how the age and mass of the object can be deduced.

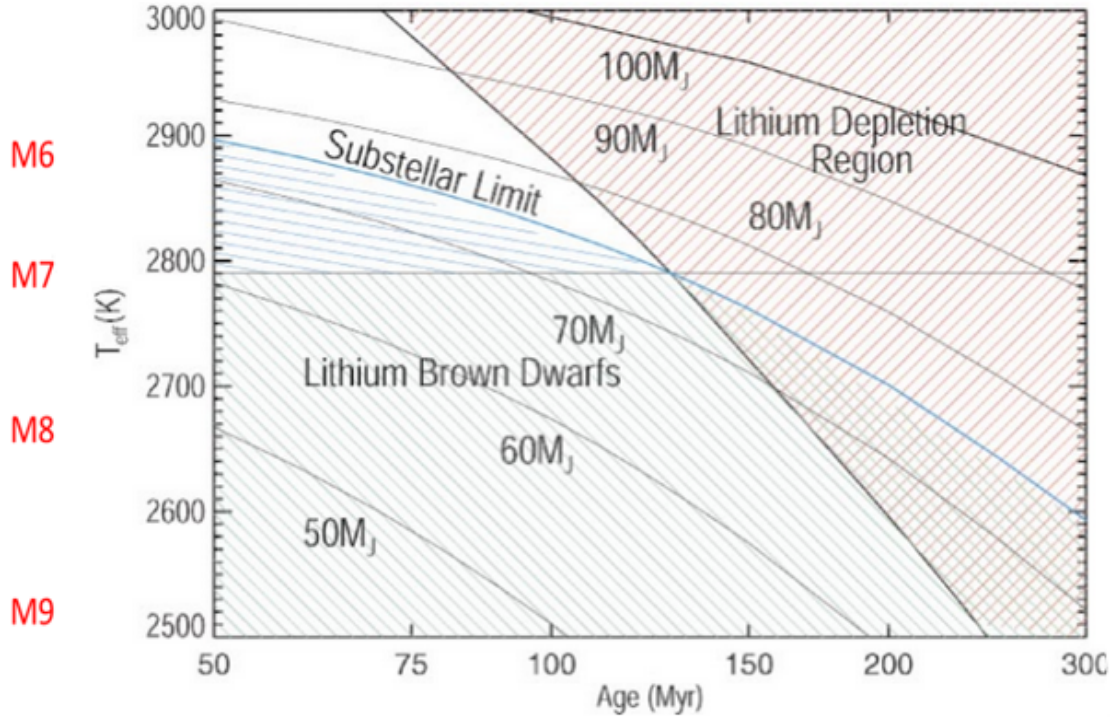


Figure 1: Lithium-Temperature Constraint.

The more-or-less diagonal curves are the tracks along which low-mass stars and brown dwarfs of the labelled mass evolve (getting cooler with age). The red region shows the area of the temperature versus age plot where Li is expected to be depleted. The green region marks where the presence of Li guarantees that you have a brown dwarf with mass $< 0.075 M_{\odot}$. The sub-stellar limit at $75 M_J$ is noted in blue. The horizontal line marks the temperature at which the sub-stellar boundary crosses the depletion region. The red/green area is that of false positives, while in the blue dotted region it does not distinguish between stars and brown dwarfs (unless their age is known). In fact, it is clear that if we observe an extremely old object but still with the presence of lithium, it is probably a brown dwarf (falling into the green zone), as the lithium is not burned. However, if it is a very young object, we cannot say anything. To give some examples: an M6.5 (2850 K) object with Li could either be a brown dwarf or a star younger than about 140 Myr, but an M6.5 object without Li is a star. An M8 (2650 K) dwarf with Li is a brown dwarf with an age less than 200 Myr, but an M8 dwarf without Li could either be a more massive brown dwarf with age just over 200 Myr or an older low-mass star. However this test may fail at much cooler temperatures (cooler than about L8) because the Li atoms form LiCl and LiOH molecules, the atomic Li-I line at 670 nm disappears from the spectrum, and the molecular features are too weak/confused to be identified. To sum up: an object with a spectral type of M8 or cooler with Li is a sub-stellar brown dwarf but object with Li but a warmer spectral type must be a brown dwarf if it is younger than some threshold age (e.g. about 50 Myr for type M6). Otherwise, an object younger than ~ 150 Myr without Li must be a star but an object cooler than about spectral type L2 must be a brown dwarf whether it has Li or not. The next question is therefore where to apply the lithium test. Below we will look for, young associations and moving groups, young clusters and star forming regions.

2.1 Young associations and Moving Groups

Given their proximity and youth, they represent the best targets for detecting and resolving sub-stellar companions and circumstellar disks. Furthermore, because their spectra are relatively sharp ($A_V < 1$), members of the closest associations are good candidates for lithium testing and spectral classification for young stars and brown dwarfs. Typically two types of strategies are used to detect low-mass stars and brown dwarfs within nearby stellar groups and associations: Photometric and astrometric data (that is, an analysis of data on the brightness, color and movement of stars) and control of late-type dwarfs found during field surveys (it examines already detected late-type stars to determine whether they show signs of youth or whether they share space motion with known members of nearby stellar associations).

2.2 Young clusters

Open clusters are more compact in the sky, so they can be photographed in their entirety. The identification of candidate members is usually based on proper motions and/or optical color-magnitude diagrams. The problem is that an object might look like a brown dwarf because it displays lithium, but it might actually be a young star that has not yet used up all its lithium (giving the possibility of a false positive in the test). Or, if we wait long enough, even high-mass BDs will deplete their lithium (giving the possibility of a false negative in the test). Furthermore, if we look at regions of star formation, it is difficult to distinguish stars from brown dwarfs, due to the intense star formation. For example PPL 15 is an object only slightly fainter than HHJ 3 (another Pleiades member) but lithium appeared only in PPL 15. These

problems actually constrain the age of a young cluster, as happened to the Pleiades that with an age of 115 Myr (rather than the classical age of 70 Myr), the behavior of the objects makes sense. The lithium test opens the door to a new method for quantifying the age of open clusters (i.e. the youngest ones).

2.3 Star-forming region

Embedded protostars are more difficult to identify and study due to the extinction of dust and molecular clouds. Because of the extreme youth of these objects, they can be identified via signatures of activity (X-rays and UV), accretion ($H\alpha$, UV), and disks (N/mid-IR). As in open clusters, candidate low-mass members of star-forming regions also can be identified with proper motions and color-magnitude diagrams measured from optical and near-IR images. However proper motion selection of candidates has not been widely used in star-forming regions since extinction prevents many of the members from appearing in photographic plates. Since clusters with ages of > 3 Myr usually have little extinction ($A_V < 2$), standard color-magnitude diagrams like those used for open clusters are sufficient for the selection of candidates. However, extinction from the parent molecular cloud in younger regions can be quite large, and it varies significantly from one line of sight to another ($A_V = 0-50$), which increases the contamination of reddened background stars in the area of a color-magnitude diagram inhabited by cluster members.

3 Classes and types of brown dwarfs

Typically, M stars are characterized by molecular lines of TiO (titanium oxide) and VO (vanadium oxide) in the optical and NIR. However, some spectra have been observed shifted to the red (therefore very cold) but without the characteristics of the spectrum of M-type stars. This pushed astronomers to define a new class of objects, namely the L class. Not all L stars are BDs, nor are all BDs L stars. Whether or not a BD is an L star depends on both its mass and its age. The temperature is between 1200 and 2000 K and they typically emit in the infrared. What distinguishes L stars from M stars is that they are so cool that Ti has been captured in refractory grains, and is not visible in the red in molecular bands. Furthermore, spectra with broad potassium lines, hydrides, presence of lithium and lack of TiO band are the main characteristics of an L-band object. The only atomic features visible in the optics, for VLMS, are lines of neutral alkali metals, such as Na and K, as well as the much rarer C and Rb (and of course sometimes Li). In the range between 650-900 nm, the most striking is the K resonance doublet at 766-770 nm, which merge together and become a very broad bowl-shaped feature covering more than 10 nm (as can be observed in the Figure (2)). While in the near infrared, the vapor bands become stronger and stronger, along with H_2 and CO. Also dust plays a fundamental role in a star's spectra. It creates absorption, diffusion, diffraction and emission effects. Methane has also been observed in BD atmospheres. Indeed at extremely cool temperatures ($< 1,400$ K, which only BDs can attain), methane becomes an increasingly important form for carbon molecules. In fact, most BDs in the Galaxy should be methane dwarfs, since they will cool to the required low temperatures within 1 – 2 Gyr. This type of dwarf is placed in an even different class (class T), that is, a class in which their spectrum is dominated by methane lines. More massive the object is, more it is able to capture light particles. For example, Jupiter's atmosphere is completely different from Earth's, in fact we mainly observe hydrogen (which,

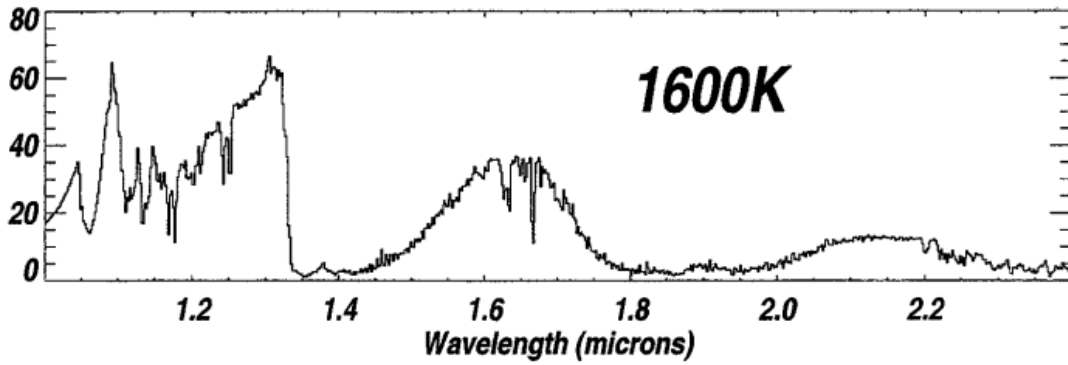


Figure 2: Infrared model spectra of very cool stars. The three humps at 1.2, 1.7, and 2.2 μm are caused by water absorption in the objects.

being lighter, requires a greater gravitational field to retain it). The same is true for brown dwarfs. Furthermore, it must always be considered that the atmosphere is opaque at certain wavelengths but transparent at others. This limits the detection of the brown dwarf.

3.1 Multiplicity of brown dwarfs

Multiplicity refers to the phenomenon where two or more stars are found in a multiple star system. Studying multiplicity can provide valuable information on the formation and early evolution of low-mass stars and brown dwarfs. Before 1995, the lack of success in imaging or radial velocity surveys for brown dwarf companions to nearby stars led to pessimism, termed the "*brown dwarf desert*." However, subsequent searches around various types of primary stars revealed a low fraction ($<1\%$) of stars harboring well-separated visible brown dwarf companions. However, if you look for binary systems composed of both BDs, the fraction of these systems is much more likely. Why? Why is there a lack of BD around higher mass primaries?

To answer these questions, research is aimed at the formation and evolution of such systems. This will be discussed in Sect. (4) and Sect. (5).

3.2 Characteristics of young stars and brown dwarf

After a young low-mass star or brown dwarf is discovered, its stellar properties are studied, such as spectral type (see Sect. (3)), effective temperature, extinction, bolometric luminosity, mass and age.

Effective temperature: The effective temperatures of young low-mass objects were estimated via two main methods: conversion of spectral types and fitting of absorption lines into their spectra. A temperature scale has been developed to align the isochronal ages of multiple GG Tau systems. Furthermore, temperatures were derived from atmospheric models via high-resolution spectral line fitting. These approaches offer a complete picture of the thermal properties of young stellar objects.

Extinction: Young brown dwarfs in the solar neighborhood have negligible extinction ($A_V < 0.5$), but members of star-forming regions are often obscured by their natal molecular clouds. For quantify the redemption from extinction it's possible define the ratio between extinction and color excess, indicated by $R \equiv \frac{A_V}{E(B-V)} \approx 3$, where depending on the region being observed,

it has an approximately constant value. For example for our galaxy $2.5 \lesssim R \lesssim 5$.

Bolometric luminosity: Bolometric luminosity is a measurement of the total amount of energy emitted by a celestial object in all wavelengths of the electromagnetic spectrum. However many young low-mass stars and brown dwarfs lack photometry across a wide enough range of wavelengths for direct calculations of their bolometric luminosities. With new technology N/mid-IR data are now available for a sufficiently large number of young low-mass objects that one could estimate the typical bolometric corrections as a function of spectral type at young ages.

Mass and age: Masses and ages for young low-mass stars and brown dwarfs are normally estimated by comparing their temperatures and luminosities to the values predicted by theoretical evolutionary models.

4 The Initial Mass Function (IMF)

How do brown dwarfs form? The goal now is to create a "*universal*" sub-stellar mass function, in order to understand under which conditions brown dwarfs form. There're several theory about the formation of brown dwarf, like gravitational compression and fragmentation of gas in a massive collapsing core or ejection phenomena. Also photoionizing radiation from O-B stars and gravitational fragmentation and turbulent compression of massive circumstellar disks around stars can also be causes of formation. But which of these is the dominant process?

In this section, we summarize IMF (Initial Mass Function) measurements for low-mass stars and brown dwarfs near the Sun and in young clusters, and discuss their impact on the formation of such objects. Clearly, several errors can be made in evolutionary models for IMF. For IMFs in the younger clusters, the masses are derived from the positions in the H-R diagram (see below) and therefore there is no dependence on luminosity. However, the spectral types of young low-mass stars and brown dwarfs could be altered by magnetic activity. To evaluate the magnitude of this error, one can consider the width of the peak in the distribution of spectral types for a star-forming region. If the peaks are narrow, it indicates that most low-mass members are not experiencing significant changes in spectral types.

4.1 Solar neighborhood and galactic disk

Measurements of the IMF are frequently characterized in terms of the power-law forms $dN/dM \propto M^{-\alpha}$. The slope **for stars** was found to be approximately $\alpha \approx 2.35$ until $0.5M_{\odot}$, then flattening out with $\Delta\alpha \approx 1.5$. While the sub-stellar (brown dwarf) mass function in the solar neighborhood shows a slope of $\alpha \approx 0$. However, as we will see for young clusters, $\alpha \approx 0.5$ is found, but this difference is not significant since the sub-stellar mass functions in the field and in the clusters have been derived by different methods. There is no evidence of the surplus of brown dwarfs in the field compared to young clusters, which is expected if brown dwarfs are born with much higher space velocities than stars. Brown dwarfs at $T_{eff} \lesssim 300$ K are particularly valuable since their numbers are sensitive to minimum masses of the IMF in the range of $5 - 10 M_J$.

4.2 Young cluster

Regarding the mass function for young clusters, it is expected that brown dwarfs, being less gravitationally bound, often concentrate in the peripheral regions of star clusters, where they are more susceptible to being ejected first by dynamical interactions, suggesting that they may be among the first to move away from the cluster. The sub-stellar mass function inferred from the Pleiades is gently rising. We can characterize it as done for the solar neighborhood, with the index α in the equation $dN/dM = M^{-\alpha}$. It appears that this index has a value of about $\alpha \approx 0.5$ for this cluster. Of course, one never observes the MF directly, but rather the luminosity function and then, with theoretical models, allow the conversion to the MF. Under comparison analyses, it was possible to compare also several young clusters. It was found that the only significant change in the low-mass IMF is the unusually high peak mass at Taurus. Figure (3) shows a comparison of spectral type distributions for representative samples of members of Taurus, IC 348, and Chamaeleon I. The distributions in IC 348 and Chamaeleon I peak at M5 while Taurus exhibits a large surplus of K7-M1 stars ($\sim 0.8 M_{\odot}$). Previous explanations for

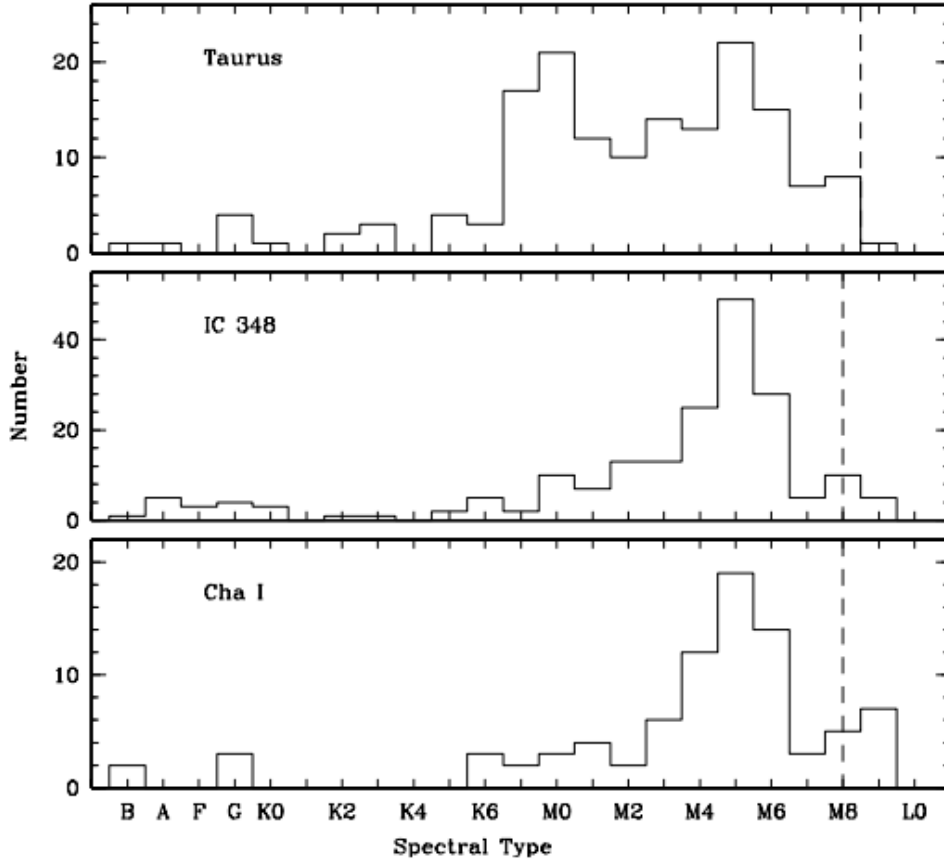


Figure 3: Distributions of spectral types for representative samples of members of Taurus ($\tau \sim 1$ Myr), IC 348 ($\tau \sim 2-3$ Myr), and Chamaeleon I ($\tau \sim 2-3$ Myr) [1].

the unusually high peak mass in Taurus have involved a higher average Jeans mass compared to other regions. Since the mass function of brown dwarfs is roughly the same regardless of the presence of O stars, it's possible to conclude that halting of accretion by photoionizing radiation isn't the dominant process that results in the formation of brown dwarfs. Similarly, Taurus has a very low stellar density, and yet it has produced a comparable abundance of brown

dwarfs as much denser clusters like the Trapezium, indicating that dynamical interactions are not essential for the formation of brown dwarfs. If brown dwarfs form predominantly in disks around solar-type stars, then the numbers of brown dwarfs and solar-type stars should scale together, but this is not seen in the comparison of Taurus to other clusters. Finally, we note that the mass functions of pre-stellar cores in molecular clouds resemble the stellar IMF in terms of the Salpeter slope at high masses and the flattening at low masses, which has been taken as evidence that gravitational or turbulent fragmentation determines the masses of stars, and perhaps brown dwarfs.

4.3 Binary system and field BDs

There are several methods to identify a BD around a primary star, for example by looking radial velocity of the primaries (astrometry) or Doppler effect. The difficulty with these candidates is precisely that they have lower mass limits, and are difficult to detect. It has been observed that, as $q \equiv M_2/M_1$ increases, i.e. when the mass of the second component is relatively greater than the first, binary systems become less common. High-resolution imaging of the members of Taurus and Chamaeleon I has found that the frequency of wide binaries and their average separation decrease steadily from solar-mass stars to brown dwarfs. It's possible to compare those predictions to observations, which can be summarized as follow: brown dwarfs have a binary fraction of 20–30% most of the binaries are tight (<20 AU) and a few are very wide (>100 AU), and the mass ratios tend to approach unity. However, it remains unclear whether any of these models can make wide binary brown dwarfs in the low density conditions in which they have been primarily found.

This result should not be surprising since, where it was possible to create a star (therefore with temperatures capable of triggering hydrogen fusion), it is plausible that the same mechanism occurred in nearby regions too. This is why seeing a BD in a binary system with a primary is extremely rare. A second explanation is also the tidal interaction between a massive (primary) object with a BD. This could cause partial destruction of the BD. It is more likely to find a brown dwarf binary (an example is Cwise J014611). The mass function for binary systems is certainly much more complicated, due to the low statistics. Even for field BDs the analysis to obtain a sub-stellar MF is rather complicated. Field investigations require two distinct and difficult phases: the first is correcting the investigation for observational biases and effects, the second problem lies in the nature of the BDs themselves. By definition, they never enter the main sequence and therefore continually fade away with time. Observations support a mass function with α less than 2 (suggest 1.3). This implies a slightly larger number of BDs than the cluster result. Such a mass function means that BDs aren't a dynamically important mass constituent of the disk and are unlikely to contribute significantly to baryonic dark matter (this would require α greater than 3). The spatial density of BDs is 0.1 systems per cubic parsec. The total number of BDs could therefore easily exceed the total number of stars. So, binary companions (especially of solar-type stars) are not a good means of addressing the general sub-stellar. They probably tell us more about the binary formation mechanism.

4.4 Circumstellar disks and protostars

As mentioned above, stars are born from a contraction of molecular clouds. These clouds are rotating and since angular momentum is conserved, the cloud flattens forming a circumstellar disk. It has been proposed that brown dwarfs arise when this accretion is prematurely halted.

The impact of falling material on the stellar produces an accretion shock, which radiates at UV and optical wavelengths. Since $H\alpha$ is more easily observed in brown dwarfs than UV emission, line profiles of $H\alpha$ provided the first evidence of accretion for young low-mass objects. It is possible to demonstrate that the (spherical) growth rate is $\dot{M} \propto M^2$. Because of their low luminosities, most low-mass stars and brown dwarfs do not heat their disks sufficiently for significant excess emission to appear at 2-3 μm . The arrival of the Spitzer Space Telescope resulted in a breakthrough in surveys for disks around brown dwarfs (as well as stars in general). In general, circumstellar disk can be observe in mm and infrared band because that is where the dust emits. In general, disks around smaller objects have a longer lifetime than disks around more massive objects. Disks have been observed around brown dwarfs and can be expected to be typically smaller (20-40 AU) than disks on stars (70-100 AU). This dampens the degeneracies between radius and other disk parameters. In general, the size of the disk can vary due to truncating effects (for example a *fly-by*), which also drastically changes the subsequent evolution of the disk. The measurements of disks around low-mass stars and brown dwarf provide valuable constraints on the stellar mass dependence of the physics of accretion, disk evolution, and planet formation, but unfortunately they have not offered direct tests of the possible formation mechanisms for brown dwarfs.

Brown dwarf formation mechanisms arise during cloud fragmentation or during the protostellar phase, offering direct constraints on formation patterns via observations of their youngest progenitors. Brown dwarfs formed by disk fragmentation or ejection should not appear as isolated protostars, characterized by the presence of a growth disk. To date, Spitzer surveys have identified more than a dozen candidate class 0¹ sources with luminosities that are indicative of young brown dwarfs.

5 Early stellar evolution at low masses

The early evolution of low-mass stars and brown dwarfs ($\tau \lesssim 100$ Myr) can be characterized largely in terms of changes in bolometric luminosity, effective temperature, radius, and angular momentum over time for a given mass stellar. Structures as circumstellar disk or wind indicate the evolution of the angular momentum, while there are theoretical models that have made predictions for the evolution of the first three parameters. The aim in this case would be to reconstruct an HR diagram and understand how brown dwarfs and low-mass stars evolve over time. From the measurements of rotation, dynamic masses and HR diagram it is possible to test the theoretical predictions of the models for the evolution of the parameters mentioned above.

5.1 Rotation

In general, the faster the rotation of an object, the higher its magnetic activity, leading to emission in spectral lines such as $H\alpha$. This in turn leads to a magnetized wind from the object that carries away angular momentum and spins it down (reducing the level of activity). The field is generated by the convective movements present inside (dynamo effect). To understand the intensity and rotation of the object, we rely mainly on the $H\alpha$ line. Since $H\alpha$ emission is

¹The star formation phases are divided into class 0, I, II and III, with class zero indicating the initial phase, when the protostar

often associated with active magnetic processes, a star that rotates more rapidly emits more $H\alpha$. However, it seems that for brown dwarfs the traditional connection between rotation and stellar activity may be completely broken or even reversed. Some brown dwarfs and VLMS have been observed to rotate rapidly but show weak or no $H\alpha$ emission. This suggests that, despite their high rotation speed, these stars may not develop intense magnetic activity or active magnetic processes associated with $H\alpha$ emission as in M-class stars. This problem still has no explanation, but many theories attribute this problem to the object's atmosphere.

It is possible to track angular momentum via periods of rotation, measured by photometric analysis. For masses of $0.4 - 1.5M_{\odot}$, the periods range from $0.6 - 20$ days and have a bimodal distribution while stars at lower masses rotate more rapidly. For young brown dwarfs periods is $\lesssim 1$ day. To explain the presence of slowly rotating stars at ages of only a few Myr, it has been proposed that a young star can lose angular momentum by transferring it to circumstellar disk. If this mechanism is valid, then slow rotators should show a higher frequency of disks. The disk-locking theory has been verified for slowly rotating solar stars, but not for lower-mass stars. For lower mass stars and brown dwarfs, the angular momentum may remain approximately constant over time, or decrease only slightly due to so-called "*wind-breaking*", a process in which the stellar wind gradually reduces the rotation rate .

5.2 Dynamical masses

"*Dynamic mass measurements*" refer to estimates of stellar masses obtained through analysis of the orbital motions of binary or multiple star systems. Dynamic masses are independent of theoretical models and provide a direct estimate of the mass of the stars involved. Theoretical predictions are compared with the observed properties of stars and brown dwarfs for which dynamical measurements are available. Radii and temperature are quite in agreement with theoretical models, however it was discovered that for the binary system 2M 0535-0546, one of the components of the binary system is hotter than predicted by standard theoretical models. These discrepancies can be attributed to factors such as the presence of strong magnetic fields or stellar activity.

5.3 H-R diagram

To display the early evolution of low-mass stars and brown dwarfs, we plot objects on a diagram of M_{K_s} versus spectral type. The graph is shown in the Figure (4). First of all, we note that the sequences for Taurus, Chamaeleon, and U-Sco have widths of 1-2 mag in M_{K_s} at a given spectral type while the Pleiades sequence is much more narrow. Cluster sequences steadily converge from earlier to later spectral types, finally approaching the sequence for field dwarfs. From brown dwarfs in Taurus to brown dwarfs in the Pleiades, the sequences lose about 3.5 mag when moving from M7 to L0. This is consistent with expected brightness changes over the 1 to 100 million year timescale for brown dwarfs with temperatures of 2900 and 2200 K, suggesting that there is a significant reduction in the brightness of brown dwarfs as they transition from state of youth at maturity. We also note that the luminosities of the young brown dwarfs (younger than M9) in the diagram are on the field dwarf sequence or even below it, which is unexpected since young brown dwarfs should be brighter than the younger ones. For example 2M 1207-3932 B and companions to HR 8799 appear below the theoretical isochrones predicted for their estimated ages. This under luminosity phenomenon appears to be common to many young

brown dwarfs and is attributed to several causes, including a possible temperature dependence on the gravity of the L/T transition and the planetary nature of the companions.

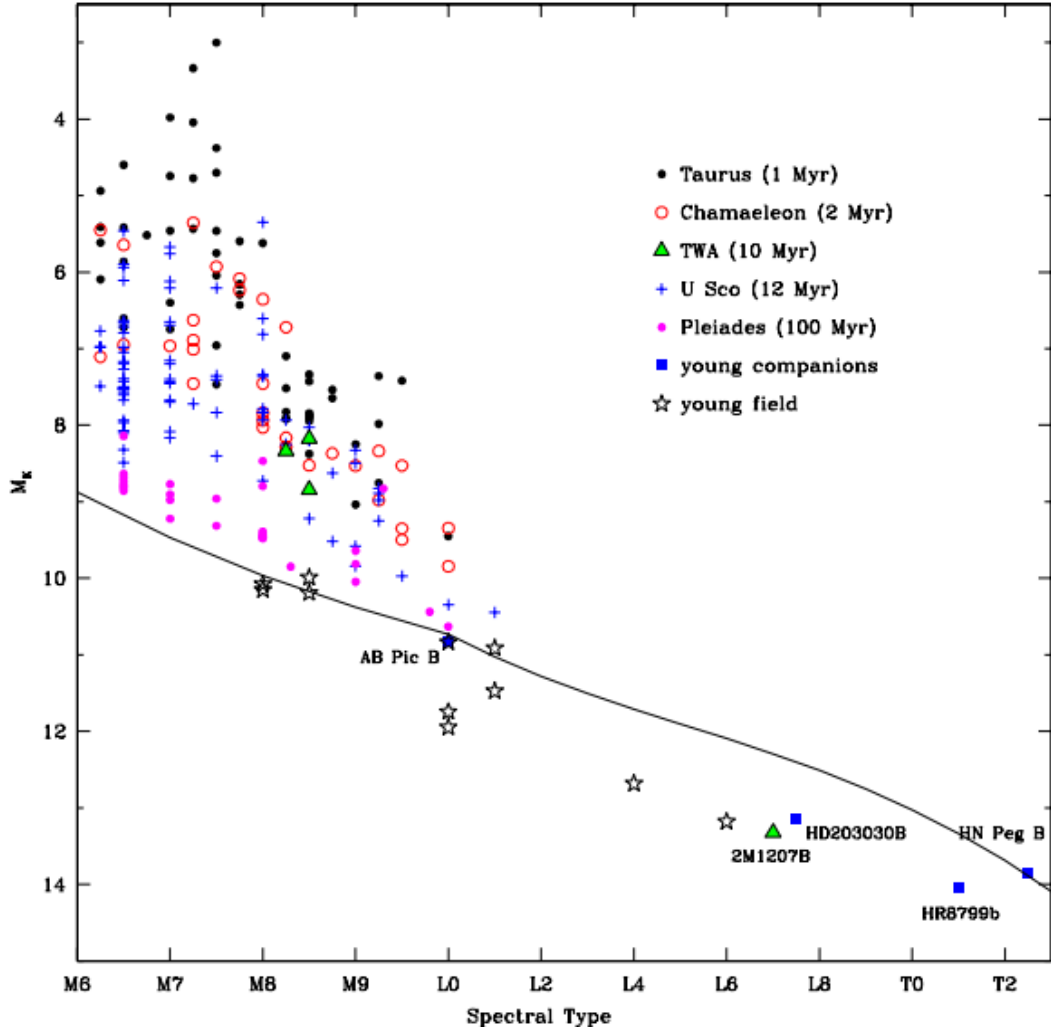


Figure 4: M_K versus spectral type for young late-type objects in nearby associations, clusters, multiple systems, and the field. A fit to data for normal, older field dwarfs is shown.

6 Conclusion

The absence of a significant dependence of the abundance of brown dwarfs on stellar density, the ability of brown dwarfs to form in isolation and in wide binaries, and the likely existence of protostellar brown dwarfs together provide compelling evidence that brown dwarfs can form without the involvement of tidal shear in massive cluster-forming cores, dynamical interactions, disk fragmentation around solar-type stars, or photoionizing radiation. Thus, it seems likely that the one remaining formation mechanism is turbulent fragmentation.

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